

Anti-money laundering systems: a systematic literature review

Anti-money
laundering
systems

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Abstract

Purpose – This paper aims to understand and document the state of the art in the anti-money laundering (AML) systems literature.

Design/methodology/approach – A systematic literature review (SLR) is performed using the Saudi Digital Library. The outputs published as conference proceedings, workshop proceedings, journal articles and books were all considered. The final sample size after omitting out-of-scope selections was 27 documents, which mainly span from 2015 to 2020.

Findings – The sample is discussed based on a categorization, which demarcates solutions, machine learning, data sources, evaluation methods, implementation tools, sampling techniques and regions of study.

Originality/value – This SLR could serve as a useful basis for researchers and salient decision-makers, who are seeking to understand the nature and extent of the currently available research into AML systems.

Keywords Algorithms, Anti-money laundering systems, Detection, Machine learning, Systematic literature review

Paper type Literature review

1. Introduction

Money laundering is defined as the process of hiding an illicit origin of dirty money and making it seem legitimate and valid (Le-Khac *et al.*, 2016). ML can also be defined as the process of cleaning “dirty” money, which means funds collected from criminal or illegal activities including drug trafficking, illegal gambling and tax evasion (Soltani *et al.*, 2016) and (Salehi *et al.*, 2017). Another definition of ML is the “process of converting unaccountable money into accountable money” (Suresh *et al.*, 2016).

ML has a negative impact on the global economy, and it is considered a serious problem discussed by countries around the world (Syed Mustapha Nazri *et al.*, 2019). Indeed, ML is the third largest business around the world, accounting for about 2.7% of global gross domestic product (GDP) after the currency exchange and auto industries (Le-Khac *et al.*, 2016; Soltani *et al.*, 2016). The International Monetary Fund estimated that ML proceeds are between 2% and 5% of global GDP (Syed Mustapha Nazri *et al.*, 2019). Also, as stated by the United Nations Office on Drug and Crime, the laundered money in one year amounts to between \$500bn and \$1tn across the world and of this money about \$400bn–\$450bn comes from drug trafficking (Le-Khac *et al.*, 2016). ML has three main stages: placement, layering (also known as occultation) and integration. In the first stage, funds are obtained through criminal activities and introduced into the financial system. Then, in the layering stage, the original source of the money is hidden by spreading the money to multiple intermediaries.



Finally, in the integration stage, the illegal cash is transferred to the owner (Alexandre and Balsa, 2015; Salehi *et al.*, 2017; Savage *et al.*, 2017; Soltani *et al.*, 2016; Suresh *et al.*, 2016).

The objectives of the Financial Action Task Force (FATF) group are to provide standards and enhance legal, regulatory and operational measures to fight ML, terrorist financing and any other threats that would affect the probity of the international financial system [About - Financial Action Task Force (FATF), 2020]. Generally, most governments keep looking to improve processes to minimize or prevent illegal activities that affect capital (Syed Mustapha Nazri *et al.*, 2019). In addition, governments around the world have issued regulations and recommendations to combat ML. For example, the Financial Transactions and Reports Analysis Centre in Canada is responsible for establishing regulations and policies in that country, affecting accountants, banks and real estate organizations. In the USA, FinCEN provides regulations and recommendations to financial institutions. International agencies such as the FATF also set standards and give recommendations for fighting ML (Soltani *et al.*, 2016).

The detection of ML commenced in 1970, when financial institutions started notifying huge transactions to their governments. In the late 1990s, statistical techniques were used to detect ML patterns, specifically Bayesian models and temporal sequence matching, and in 2004, machine learning techniques began being used to the same end. The most popular algorithms for detecting ML activities are the C4.5 decision tree (DT), support vector machine (SVM) and radial-based function neural network model (Soltani *et al.*, 2016). A system that works against ML is called an anti-money laundering (AML) system.

The intention herein is to conduct a systematic literature review (SLR) that mainly explores existing AML techniques, which use machine learning and other methodologies to detect suspicious transactions. Accordingly, methodological details explaining how articles were selected for analysis are provided in Section 2. The findings of the review are then documented in Section 3 in a structured way based on classifying the research outputs. The paper then culminates by offering conclusions in Section 4.

2. Methodology

2.1 Research scope and focus

The scope of this review is AML techniques and their effectiveness as well as other approaches proposed by researchers for the detection of suspicious financial activities. The main focus is on applications that use machine learning methods, social network analysis, deep learning and other AML solutions.

2.2 Research framework

A SLR was performed to survey the current state of the art of AML methods. The SLR is therefore guided by the following research question: What are the existing methods and approaches that can be used to detect ML activities?

The SLR was performed by using the Saudi Digital Library with English databases that included libraries such as Springer, IEEE Xplore, ScienceDirect, Emerald Insight and ACM. The documents in the sample mainly span from 2015 to 2020, with two items from 2010 and one from 2011. The keywords used while searching in the digital library were as follows: *Anti-money laundering*, *money laundering*, *detecting money laundering*, *AML*, *Anti-money laundering systems*. The final sample size after omitting out-of-scope selections was 27 documents.

2.3 Outputs: type and temporality

The outputs published as conference proceedings, workshop proceedings, journal articles and books were all considered. Patent applications, lecture notes/slides, software products and information websites were not within the sampling frame. [Figure 1](#) shows the temporal distribution of the sample in terms of when outputs were published.

3. Results: classification of research outputs

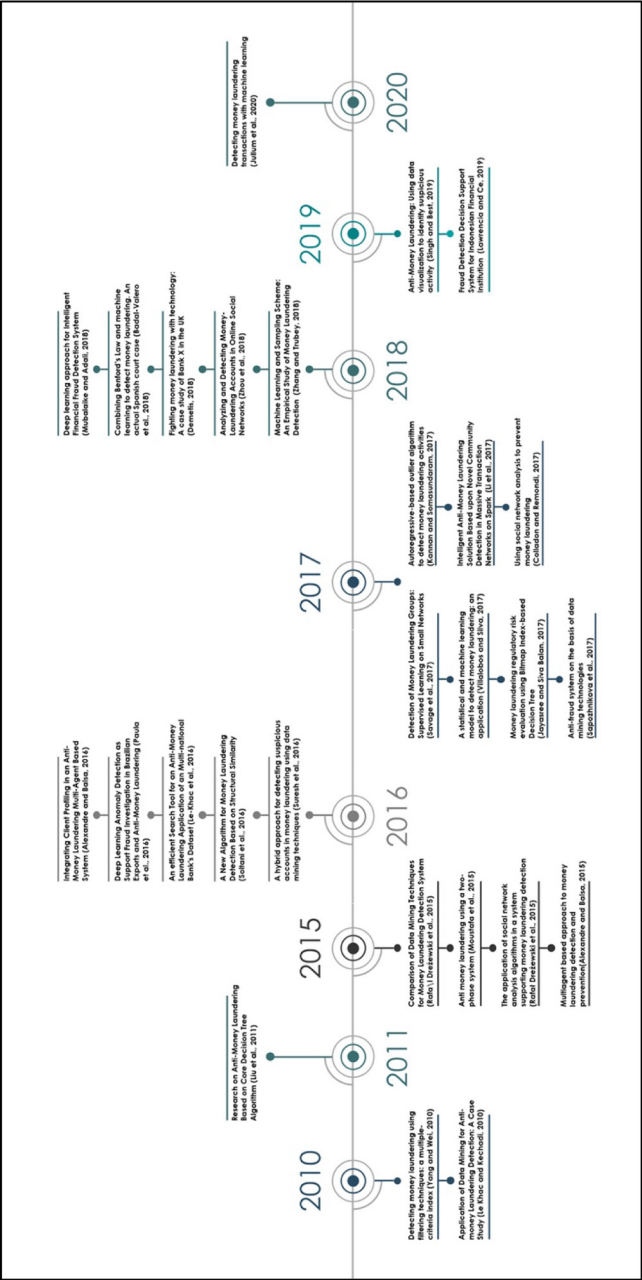
3.1 Anti-money laundering solutions

This section presents AML systems that use new frameworks or any other techniques not encapsulated under machine learning or deep learning. To start, in a study by [Soltani et al. \(2016\)](#), a novel framework is proposed which is designed to take a set of bank account transactions and then return a list of suspicious activities. The framework commences by reviewing the input data and looking for similarities between transactions with common attributes (e.g. similar deposit and withdrawal amounts). It then generates a graphical representation of all similar transactions. Next, a network-based algorithm is used to refine accounts and transactions based on *balance score*. The framework uses a clustering technique to detect suspicious groups. Lastly, the suspicious communities would be rearranged and ordered as the framework's output.

In addition, an AML solution is developed by [Le-Khac et al. \(2016\)](#) for application with respect to banks' databases. The authors developed a search tool and implemented Web services using an index tree, creating a system targeting both end users and administrative users. Their solution helps to overcome data quality issues and provides a global view of all customer information without changing the current architecture of banks' databases. Data from banks and the National Court Register were used to build and analyze social networks to investigate ML cases. The clustering techniques and frequent pattern mining were used to detect the roles of people involved in criminal activities, assignment of roles allowed for identification of true bosses and vulnerabilities within the networks. The Money Laundering Detection System (MLDS) is used to support police analysts and this was developed in collaboration with the Polish National Police ([Rafał Dreżewski et al., 2015a](#)).

A plan-based framework for AML systems has also been developed and demarcated into two phases. First, the monitoring phase is used to detect potential laundering and depends on several methods, namely, rule base, feature extraction, clustering and cycle detection. Second, the STRIPS-based phase is used to support beliefs in suspicious transactions and is presented in a study by [Moustafa et al. \(2015\)](#). Moreover, visualization techniques are used in a study by [Singh and Best \(2019\)](#) to identify ML activities by showing how the link analysis visualizes bank transactions and detects suspicious activities. The development of the prototype had four phases: task analysis, which identified the problem to be solved; system design, which planned the specifications for data, processes and interfaces; implementation, which aimed to operationalize the system as a proof of concept; and testing, which evaluated the system in terms of whether it fulfilled its purpose. The prototype is called AML²ink. Next, [Kannan and Somasundaram \(2017\)](#) proposed a triangular boundary-based outlier detection (TBOD) and autoregressive-based outlier algorithm for money laundering detection (AROMLD) using a data set of online bank transactions. Using this method, outliers are detected based on two different models, using TBOD and AROMLD techniques. With the TBOD technique, the profiles of customers are subjected to a triangular area map created to find correlations between transactional details by segmenting the data

Figure 1.
Published articles in
money laundering
field with years of
publication



into groups based on customers' risk and product use. The alternative technique was AROMLD, which included mean, zero mean, autoregression and inter quartile range (IQR) operations; these operations would first be used for detecting ML activities upon extraction of the data set, then for distinguishing between normal and outliers' transactions, with the regression deviation-based IQR defining the threshold. Thus, if the regression value is more than the threshold, it would be considered an outlier. If not, the value would be normal. In addition, [Yang and Wei \(2010\)](#) developed a multiple detection approach (MDA) that focused on evidence given by data to detect ML stages (placement, layering and integration). The MDA is based on three models. The first is a transaction assessment, which considers volume and recurrence through an outlier detection model. The second model was a trading correlation test that aimed to detect unusual trades or transactions made by a company. The final model aimed to detect fraud of a financial nature by applying Benford's law. The results from all three models are then combined to end up with a span from zero (no ML suspicion) to three (high ML suspicion).

Moving on, in [Demetis \(2018\)](#), an empirical contribution is put forward for refining the true positive rate (TPR) of transaction monitoring systems based on data for a case-study bank; AML evaluation measures are then identified for use by the decision-makers of AML systems. The paper also put forward a conceptual framework for ML detection. [Li et al. \(2017\)](#) proposed a method for revealing suspicious activities from huge transaction networks by using a temporal-directed Louvain algorithm optimized on a Spark GraphX platform. Thus, cases with high-risk scores would be considered abnormal activity. Furthermore, [Colladon and Remondi \(2017\)](#) developed a new approach aiming to assess the risk of clients' profiles by presenting predictive models using social network metrics (SNA); the approach enabled visual analysis of the existing hidden relationships between companies that shared the same owner. Thus, the researchers concluded that risk profiles could be predicted by this method.

3.2 Machine learning: supervised learning

The goal of supervised learning has been identified by [Friedman et al. \(2001\)](#) as "to predict the value of an outcome measure based on a number of input measures." It is also known as "learning with a teacher." Along this line, [Savage et al. \(2017\)](#) represented an end to end solution by building an automated system to detect ML by analyzing group behavior in financial transactions. The system uses both network analysis and supervised learning [through an SVM and random forest (RF) classifiers] and it is designed to be used at the Australian Transaction Reports and Analysis Centre (AUSTRAC). Similarly, in a study by [Zhang and Trubey \(2018\)](#), an empirical study is put forward using machine learning to detect rare events and ML activities. Those authors studied five algorithms: DT, RF, SVM, artificial neural network and Bayes' logistic regression (BLR). Moreover, they used a regression model [maximum likelihood logistic regression (MLLR)] as a benchmark. Further, [Villalobos and Silva \(2017\)](#) developed a classification model to detect suspicious transactions, which is based on five different classification models as follows: SVM, C5.0, C&RT, neural network and chi-squared automatic interaction detector (CHAID). The best results came from C5.0.

[Zhou et al. \(2018\)](#) proposed a detection method that aimed to integrate features using a statistical classifier, such as an SVM, RF or logistic regression (LR). In addition, three classifiers were used in a transactional monitoring system, specifically a multilayer perceptron, SVM and RF ([Sapozhnikova et al., 2017](#)). Moreover, the proposed method in a study by [Badal-Valero et al. \(2018\)](#), which combines Benford's law and machine learning algorithms including LR, DT, neural networks and RF, helped to detect ML characteristics

in the context of a real Spanish court case. A decision support system was created by [Lawrenca and Ce \(2019\)](#) to support financial institutions in detecting suspicious transactions. Thus, the end result is displayed in a user-friendly dashboard form. The classification depends on fraud scenarios and uses three different methods:

- (1) a SQL statement to check the transactional total amount;
- (2) one-tail normal distribution to identify outliers and then detect suspicious transactions by SQL aggregate functions; and
- (3) a robust regression to find outliers within the linear fit.

On the other hand, [Jayasree and Siva Balan \(2017\)](#) designed a method called the bitmap index-based decision tree (BIDT) to evaluate financial ML risks. A bitmap index is defined as a data structure for efficient access to larger bank databases. The aim of indexing in BIDT is to provide indicators to the table's rows, and the purpose of the DT method is to divide the decision into smaller units for analyzing ML activity.

[Jullum et al. \(2020\)](#) developed a supervised machine learning model to control financial transactions to detect money laundering activities to differentiate between reported and non-reported transactions in DNB Norway bank. The bank has three main stages: the alert stage, case stage and reporting stage to detect illegal transactions. They used gradient boosting framework with tree model. Further, they used ten-fold cross validation in training phase to fit model and used local and iterative grid search to select hyper parameters. Meanwhile, the researchers tried to use ensemble XGBoost model with RF, Glnet or with other GBoost configuration, but it was rejected because of its complexity.

3.3 Machine learning: unsupervised learning

Unsupervised learning or “learning without a teacher” is defined as follows by [Friedman et al. \(2001\)](#): “There is no outcome measure, and the goal is to describe the associations and patterns among a set of input measures.”

To overcome ML, suggested a method that used a hash-based technique over an Apriori algorithm (association mining) and then identified suspicious activities by a graphical theoretic approach. [Le Khac and Kechadi \(2010\)](#) offered a knowledge-based solution for detecting suspicious transactions. This approach has two main elements: analysis and investigation. A clustering technique applying K-means would be used during analysis and the result of this step would be divided into two sets: suspicious and unsuspicious (a few data objects being in the suspicious set and most belonging to the unsuspicious set). Then, to increase the number of data objects within the suspicious set, a genetic algorithm with single-point crossover would be used. The output would serve as an input to the neural network for training. The DT would also represent the result and then store it in the knowledge base to help ML experts make a decision in the investigation step. Meanwhile, the MLDS was developed using four algorithms: Apriori, PrefixSpan, FP-growth and Eclat. The Apriori algorithm is used to divide money transfer into transactions based on operations executed in a specific time or operations with a specific value. The PrefixSpan algorithm is used to discover sequential patterns in a database. FP-growth is considered to be a rapid method for discovering frequent items, and it is prudent to use the Eclat method when the database is intensive, as it is a superior association rule algorithm in such cases ([Rafał Dreżewski et al., 2015b](#)). Further, applied unsupervised deep learning with AutoEncoder has been documented, with principal component analysis (PCA) being used for dimensionality reduction ([Paula et al., 2016](#)). There were differences in the performance of PCA and AutoEncoder; the researchers found that PCA was computationally expensive

compared to AutoEncoder, which helps to explain why the latter is the most commonly used method for anomaly detection.

The approach used in study by [Mubalalike and Adali \(2018\)](#) for developing an intelligent financial fraud detection system is based on machine learning techniques, such as an ensemble DT and deep learning techniques, such as a stacked auto-encoder (SAE) and a restricted Boltzmann Machine (RBM). [Liu et al. \(2011\)](#) aimed to detect abnormal activities by using a DT based on balanced iterative reduction and clustering using hierarchies (BIRCH) and K-means clustering algorithms. K-means clustering was used to split n objects into k clusters, whereas BIRCH was used for hierarchical clustering fits for large data sets. The BIRCH method could not handle financial data and was not sensitive to noise; the K-means method was better in these respects, but it was too slow for use with large data sets. Because of these issues, the core DT was based on the previous clustering methods. Further, a multi-agent-based approach to transacting with ML activities helps financial institutions confront two main problems: volume and rule improvement. The system divides agents based on their roles in the process. The first agent is used to capture and report on suspicious transactions, whereas the second agent is used to analyze suspicious transactions detected by the first agent and to make a decision ([Alexandre and Balsa, 2015](#)). [Alexandre and Balsa \(2016\)](#) developed the system proposed in [Alexandre and Balsa \(2015\)](#) by using classification

Algorithms	Article
SVM	Savage et al. (2017) , Zhang and Trubey (2018) , Villalobos and Silva (2017) , Zhou et al. (2018) , Sapozhnikova et al. (2017)
RF	Savage et al. (2017) , Zhang and Trubey (2018) , Zhou et al. (2018) , Sapozhnikova et al. (2017) , Badal-Valero et al. (2018)
DT	Zhang and Trubey (2018) , Villalobos and Silva (2017) , Badal-Valero et al. (2018) , Mubalalike and Adali (2018) , Jayasree and Siva Balan (2017) , Liu et al. (2011)
Neural network	Zhang and Trubey (2018) , Villalobos and Silva (2017) , Badal-Valero et al. (2018)
BLR	Zhang and Trubey (2018)
Apriori algorithm	Suresh et al. (2016) , Rafał Dreżewski et al. (2015b)
C&RT	Villalobos and Silva (2017)
CHAID	Villalobos and Silva (2017)
LR	Zhou et al. (2018) , Zhang and Trubey (2018) , Badal-Valero et al. (2018)
Multilayer perceptron	Sapozhnikova et al. (2017)
K-means	Le Khac and Kechadi (2010) , Liu et al. (2011) , Alexandre and Balsa (2016)
Benford's law	Badal-Valero et al. (2018) , Yang and Wei (2010)
PrefixSpan, FP-growth, Eclat	Rafał Dreżewski et al. (2015b)
AutoEncoder	Mubalalike and Adali (2018) , Paula et al. (2016)
RBM	Mubalalike and Adali (2018)
Robust regression	Lawrencia and Ce (2019)
Temporal-directed Louvain algorithm	Li et al. (2017)
SNA	Colladon and Remondi (2017) , Rafał Dreżewski et al. (2015a)
Balanced iterative reducing and clustering using hierarchies (BIRCH)	Liu et al. (2011)
Gradient boosting	Jullum et al. (2020)

Table 1.
Money laundering
detection algorithms

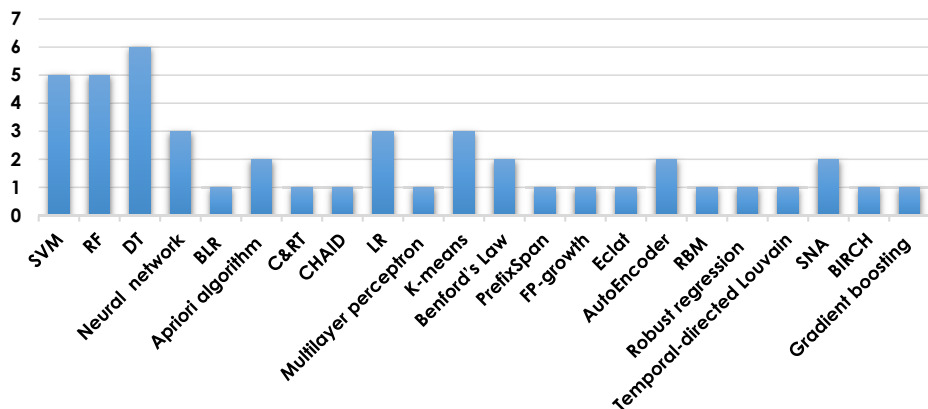


Figure 2.
Number of published
articles based on
detection algorithms

and clustering algorithms. Table 1 and Figure 2 summarize all the ML methods used by the collected articles.

3.4 Data sources

This section describes the amounts and sources of data being used by AML systems. In a study by [Savage et al. \(2017\)](#), transactional network data were collected from AUSTRAC. Similarly, [Zhang and Trubey \(2018\)](#) analyzed their results based on real transaction data, taken from a US financial institution over 13 months and hundreds of explanatory variables; they used the first eight months of data for training purposes and the remaining five for testing. [Suresh et al. \(2016\)](#) worked with a synthetic transaction data set from different banks spanning a period of 120 days and containing data on 17,000 transactions. Whereas in study by [Villalobos and Silva \(2017\)](#), data covering customers’ histories over a year were split, 50% into training and 50% into testing sets; this balanced distribution was necessary because less than 0.4% of cases were suspicious.

The data used in study by [Zhou et al. \(2018\)](#) were collected from Tencent QQ, which is one of the biggest online social networks in China and offers many services such as instant messaging, voice chat, online games and online shopping. The data set included 496,414 accounts (114,891 malicious and 381,523 benign accounts) and a cluster of 54 features to help identify the behaviors of benign and virulent accounts.

In [Sapozhnikova et al.’s \(2017\)](#) study, data were used from the UCSD-FICO database of banking transactions, created by the University of California, San Diego in 2009. The data used included more than 100,000 transactions by 74,000 customers during a period of 98 days. In [Singh and Best’s \(2019\)](#) study, data were obtained from a bank and included transactions over a two-year period from 2014 to 2016. [Le Khac and Kechadi \(2010\)](#) collected transaction data from a bank with about ten million records and 10,000 customers over the past 14 years, whereas [Badal-Valero et al. \(2018\)](#) analyzed a huge database that contained 285,774 operations among a company with a high chance of ML activities as well as a group of more than 643 suppliers. The analysis revealed 335 companies with an abnormally high number of operations, which contrasts markedly with the police expert who only identified 23 as fraudulent. The issue with the data set was that few companies were classified as fraudulent compared with the total number of suppliers in the data set.

Further, [Paula et al. \(2016\)](#) used deep learning on databases of the Secretariat of Federal Revenue of Brazil to identify exporting companies that had divergences in their 2014

exports compared to normal patterns. In [Mubalaike and Adali's \(2018\)](#) study, a data set was used based on one month of financial logs of a mobile money service company in Africa, which had more than six million transactions. In a study by [Lawrencia and Ce \(2019\)](#), data were collected from transactional logs of a multinational bank in Indonesia and from public data provided by Kaggle PaySim. [Colladon and Remondi \(2017\)](#) analyzed financial operations data related to a medium-large manufacturing company in Italy from November 2013 to June 2015. [Jayasree and Siva Balan \(2017\)](#) used a data set pertaining to Statlog German credit from the UCI repository. The data set used when developing a multi-agent system in [Alexandre and Balsa's \(2016\)](#) was from a financial institution and it described the activity of accounts through three months.

In [Jullum *et al.*'s \(2020\)](#) study, the historical data were extracted with total amount of 33,134 transactions from April 1, 2014 to December 31, 2016 from DNB bank, the set includes two types of data: customer data that randomly collected and AML alerts that were not included in the AML reports.

3.5 Evaluation methods

This section aims to demonstrate the various types of evaluation methods for proposed AML systems. What follows is summarized in [Table 2](#) and [Figure 3](#).

In [Savage *et al.*'s \(2017\)](#) study, classifier performance was calculated by using the area under the curve (AUC) for both the SVM and the RF. The SVM was associated with an AUC of 0.86, whereas that of RF was 0.92. In addition, both models had a low average recall and extremely high precision, which would lead to low rates of false positives. In short, the results showed that the system performed very well in a live environment. [Zhang and Trubey \(2018\)](#) measured model performance by AUC, ranking them in the following order based on their performance from best to worst: ANN, BLR, MLLR, RF, DT and SVM. The first four of these models had an average AUC above 0.74, whereas the latter two had an

Methods	Articles
AUC	Savage <i>et al.</i> (2017) , Zhang and Trubey (2018) , Villalobos and Silva (2017) , Zhou <i>et al.</i> (2018) , Jullum <i>et al.</i> (2020)
Recall	Savage <i>et al.</i> (2017) , Sapozhnikova <i>et al.</i> (2017)
Precision	Savage <i>et al.</i> (2017) , Mubalaike and Adali (2018)
Gini index	Villalobos and Silva (2017)
FPR	Zhou <i>et al.</i> (2018) , Jayasree and Siva Balan (2017)
TPR	Jayasree and Siva Balan (2017) , Zhou <i>et al.</i> (2018) , Villalobos and Silva (2017)
F1-Measure	Savage <i>et al.</i> (2017) , Sapozhnikova <i>et al.</i> (2017)
ROC	Badal-Valero <i>et al.</i> (2018) , Mubalaike and Adali (2018)
RMSE, Kappa	Badal-Valero <i>et al.</i> (2018)
Sensitivity	Kannan and Somasundaram (2017) , Mubalaike and Adali (2018)
Specificity	Kannan and Somasundaram (2017) , Mubalaike and Adali (2018)
Execution time	Kannan and Somasundaram (2017)
Accuracy	Soltani <i>et al.</i> (2016) , Kannan and Somasundaram (2017) , Lawrencia and Ce (2019) , Mubalaike and Adali (2018) , Alexandre and Balsa (2016)
MSE	Paula <i>et al.</i> (2016)
Confusion matrix	Mubalaike and Adali (2018) , Alexandre and Balsa (2016)
R-square	Zhang and Trubey (2018) , Colladon and Remondi (2017)
MCC	Sapozhnikova <i>et al.</i> (2017)
Brier score	Jullum <i>et al.</i> (2020)
PPP	Jullum <i>et al.</i> (2020)

Table 2.
Evaluation methods
used in money
laundering detection
systems

average AUC below 0.60, which contradicts the findings from all other extant studies, which suggested that RF and SVM performed best.

In [Soltani et al.'s \(2016\)](#) study, five tests were conducted as follows to evaluate the proposed framework and algorithm: Tests 1, 2 and 3, with 1,000 nodes representing client accounts, resulted in 300 innocent transactions and 10 ML transactions; Test 4, with 10,000 nodes, resulted in 500 innocent transactions and 100 ML transactions; and the final test, also with 10,000 nodes, resulted in 200 innocent transactions and 200 ML transactions. The results of these five tests showed the accuracy to be 90% in detection of ML communities that take illegal money and move it out to recipients and 96% in detection of ML accounts of recipients. In a study by [Villalobos and Silva \(2017\)](#), the performance of five classifiers was evaluated; the best three models were then taken forward pursuant of establishing the overall best-performing model based on the correct classification, AUC and Gini index. The best classifier was C5.0, with approximately 30% of transactions being alerted (true negative) and fewer than 1% of transactions being falsely classified.

Furthermore, to measure the effectiveness of methods, [Zhou et al. \(2018\)](#) used the detection rate (also known as the positive rate), AUC and the false positive rate (FPR). They also used ten-fold cross-validation to evaluate the classifiers, with both the SVM and the RF achieving a high detection rate, high AUC and low FPR. In [Sapozhnikova et al.'s \(2017\)](#) study, the RF classifier also performed well, achieving the best result, with ten-fold cross-validation within Matthews correlation coefficient (MCC) 0.91. The multi-layer perceptron was the second-best classifier with MCC 0.82. TBOD and AROMLD were validated in [Kannan and Somasundaram's \(2017\)](#) study with 9,000 bank transactions using sensitivity, specificity, execution time and accuracy as performance measures. The researchers performed a comparative analysis between the DT, RF, naïve Bayes, TBOD and AROMLD. The TBOD achieved higher accuracy than AROMLD, albeit with an increase in execution time, which reflects its relative complexity. Indeed, the TBOD was more accurate than all the existing approaches tested by these authors. The use of regressive formula in the AROMLD renders it less computationally complex but it still performs well as shown by the calculated accuracy level of 94.8%.

In a study by [Badal-Valero et al. \(2018\)](#), algorithms were evaluated based on the following approach. First, direct modeling without taking into account the imbalanced

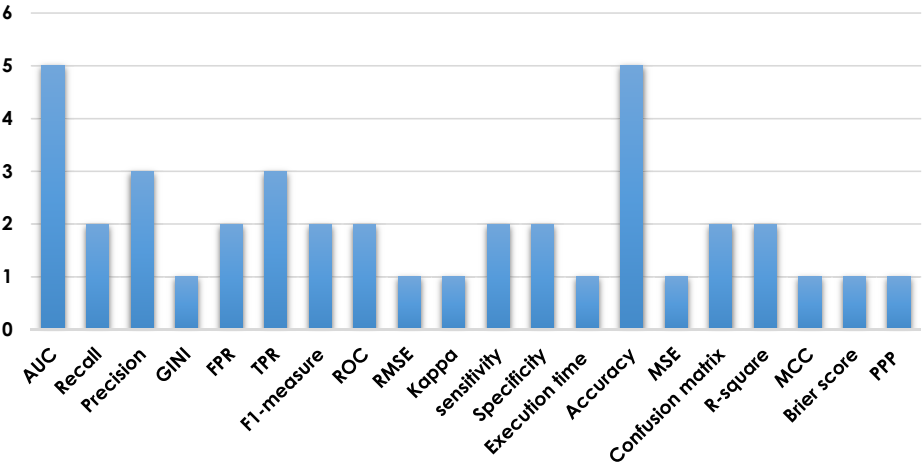


Figure 3.
Most frequent
evaluation methods
by published articles

nature of the data led to suggesting a very low true positive rate of 13.04% for the LR model, compared to 34.78% for the neural network. Given this was an imbalanced data set, the algorithms tend to prefer the dominant category and thus very few fraudulent companies were identified. Second, using a cost matrix revealed that the RF model was the only algorithm that achieved 94% correct classification of data. Third, transforming the data by using a SMOTE algorithm yielded better results than the other strategies. This strategy showed true positive rates of 82.27% for the LR model and 94.98% for the RF. In short, the neural network performed best with imbalanced data sets in terms of its receiver operating characteristic (ROC) = 0.765 and Kappa statistic = 0.340, but the smallest root mean square error (RMSE) = 0.227 was achieved by the RF algorithm. In a similar manner, the evaluation by [Mubalike and Adali \(2018\)](#) was conducted with respect to deep learning algorithms with an SAE and RBM; the end result from the confusion matrix showed that the RBM was better than the SAE and that the ROC value of the SAE was about 0.818. [Rafał Dreżewski et al. \(2015b\)](#) compared four algorithms depending on the following attributes: minimal support for frequent sequences, input file size, diversity of connections and algorithm type. First, they compared an Apriori with a PrefixSpan algorithm, with the former being shown to be slower. Next, they compared Apriori, FP-growth and Eclat algorithms. Here, the Eclat algorithm proved to be superior because it was not influenced by the number of elements whereas Apriori did not perform well because of the large number of transactions. The mean square error (MSE) was used to evaluate model performance; the greater the MSE value, the more anomalous. In short, because the system identified some ML cases that had already been identified by the domain expert, the system was considered efficient ([Paula et al., 2016](#)).

According to [Li et al. \(2017\)](#), after the internal risk management department of UnionPay conducted an investigation, it found that 9 out of 13 transactions were reconfirmed suspicious and nodes with large weights were notified to the China Anti-Money Laundering Monitoring and Analysis Center of the People's Bank of China. [Colladon and Remondi \(2017\)](#) demonstrated that working with multiple SNAs improves informative power more than a single metric. Going further, they also tested different logit models to evaluate the power of SNA metrics. The analysis showed that clients with missing information were more likely to have high-risk profiles. Results in [Jayasree and Siva Balan's \(2017\)](#) showed that the BIDT performed best based on the following measures: true positive (TP), defined as the "measure of customers correctly identified as genuine"; FP, defined as the measure of "the transaction related with a real customer who is blocked due to a name or another match, with lower FPRs correlating with more efficient methods"; the risk identification time, which is defined as "the time taken to identify the key values on money laundering accounts in the bank"; and adaptability rate, defined as the "ability of the service provider (i.e. bank) to adjust changes in services based on customers' requests during a money laundering operation."

The first phase in [Alexandre and Balsa's \(2016\)](#) study aimed to reprocess and cluster customer profiles by using the silhouette coefficient and the sum of squared error (SSE). Next, the PART algorithm was used for rule generation as well as for implementing the C4.5 DT to reveal the best leaf technique. The results of the first phase were unsatisfactory, so phase two used a new strategy with a more detailed database consisting of 2.4 million rows of customer profile data after clustering and removal of insignificant transactions. In their search to define the number of clusters, the researchers executed simple K-means, silhouette metrics, SSE, variance ratio criterion (VRC) and Van Dongen and Rand metrics. The silhouette metrics, SSE and VRC suggested six clusters, whereas Van Dongen and Rand defined seven clusters. In addition, to evaluate cluster generation, a confusion matrix with a level of accuracy was used. PAR, J48, and JRip algorithms were used for rule generation. Of

these, the J48 and JPART algorithms demonstrated the best results in the group of experiments.

In addition, in [Jullum et al.'s \(2020\)](#), there are three criteria for performance evaluation: Brier score, AUC and a new proposed performance measure by the researchers called: proportion of positive prediction (PPP) as the researchers were not able to compute metrics for the alert/case system because there is too much manual work involved and for that reason, they include PPP metric, this metric is equal to the amount of positive predictions when adjusting the TPR to a certain level. Therefore, the lower the value, the more efficient the classifier. Moreover, the AUC and Brier score were usually used to measure the performance of binary model. In addition to that, the 90% confidence interval (CI) is used for AUC and Brier score. [Table 2](#) and [Figure 3](#) summarize the evaluation methods that are mentioned in the articles.

3.6 Implementation tools

There are a variety of tools used in the implementation of AML systems, summarized in [Figure 4](#). In studies by [Savage et al. \(2017\)](#) and [Zhang and Trubey \(2018\)](#), analysis was implemented with R statistical language, whereas [Zhang and Trubey \(2018\)](#) used SAS Proc Reg for analysis. Moreover, the implementation of the proposed framework in [Soltani et al.'s \(2016\)](#) study was carried out using JAVA. [Le-Khac et al. \(2016\)](#) used C# to develop a Web-based service and SQL Server 2005. MATLAB was used for experiment implementation in [Sapozhnikova et al.'s \(2017\)](#) study. SQL queries were used to process data in [Singh and Best's \(2019\)](#) study with the GraphViz tool used to produce visualizations. SQL aggregations and Python were used in a study by [Lawrencia and Ce \(2019\)](#). [Paula et al. \(2016\)](#) used Oxdata's H2O software, which offers a set of machine learning algorithms.

In a study by [Li et al. \(2017\)](#), the proposed method was implemented on a Spark GraphX platform and an HDFS cluster was used to store all the data based on Cloudera. The JAVA platform was used in the implementation of BIDT ([Jayasree and Siva Balan, 2017](#)). The .NET platform was used in [Le Khac and Kechadi's \(2010\)](#) study for building a data warehouse using a tree topology for both customers and transactions. In addition, XGBoost library was used in [Jullum et al.'s \(2020\)](#) study. Finally, [Alexandre and Balsa \(2016\)](#) used the

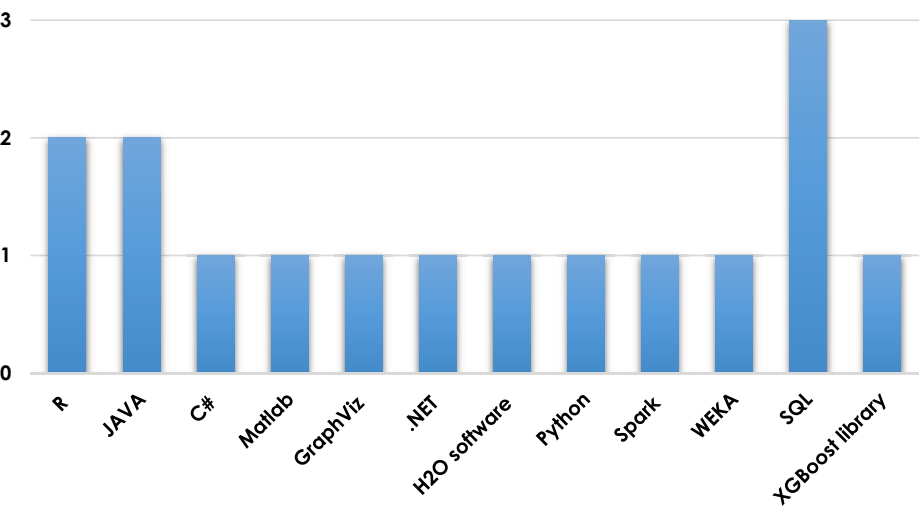


Figure 4.
Implantation tools of
published articles

Waikato Environment for Knowledge Analysis tool. Figure 4 shows a simple graph of the frequent tools.

3.7 Sampling techniques

Some articles mentioned the sampling method used to define the number of observations from the original data. Zhang and Trubey (2018) used random sampling bootstrap random sampling with replacement, as well as stratified bootstrap sampling to ensure that the derived sample contained sufficient events. As a matter of fact, the researchers started with unstratified bootstrap sampling but this did not lead to an analytically viable sample. Badal-Valero *et al.* (2018) divided their sample and made use of two different strategies: the first involved cross-validation and dividing the learning data ten-fold, then the second strategy was to divide the data 70%/30% for training and testing, respectively. In addition, Savage *et al.* (2017) used random sampling.

In Jullum *et al.*'s (2020), the data were divided into two sets: training set to train the predictive model and testing set for validation purpose and evaluation the accuracy of trained model, the training data was 28,167, whereas 4,967 for testing set.

3.8 Regions of study

This section illustrates geographical details of the sample in terms of the data used, the places from where data were extracted and the places where the AML systems were used. As shown in Figure 5, China leads the countries with four research studies in this field, followed by Poland with two research studies and the rest of colored countries have published only one research.

4. Conclusions

AML systems are categorized into supervised or unsupervised machine learning or any other techniques. Under supervised learning DT, RF and SVM algorithms are the most frequent classification algorithms used in building AML tools, whereas neural networks are one of the most frequent algorithms used in unsupervised learning. Accuracy, AUC and precision are often measured for the purposes of model evaluation. In terms of geography, the analysis found that China has the highest number of publications with four articles that mainly focused on ML crimes. In this review paper, several commercial and open-source

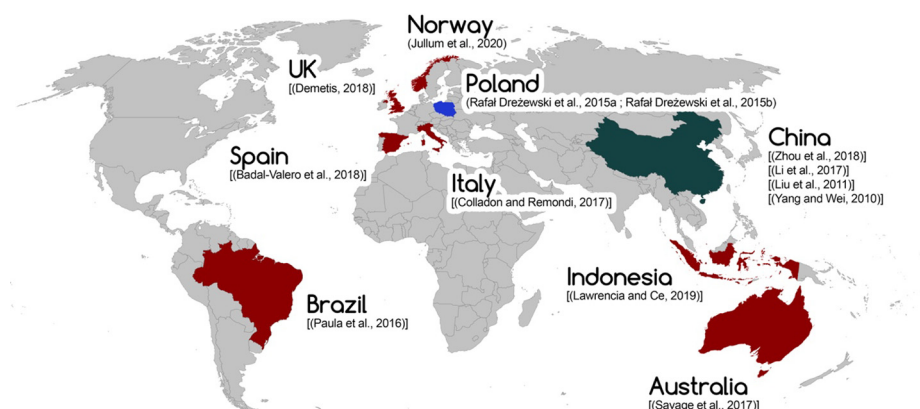


Figure 5.
Distribution of money
laundering systems

tools were found to have been used in implementation, with SQL being most frequently used followed by R statistical language and JAVA.

After reviewing the articles presented in this paper and studied them from several aspects such as: the algorithm used in AML system, the nature of data used, sampling method and finally the targeted region. The researchers found that most of the data were from banks and the current focus is on customer and financial transactions, it is better to cover other type of data from different parties, as ML is a combination of several crimes and the global experiences prove that the perpetrators of ML crimes exploit the gaps between the administrative authorities to pass the crime and give legitimacy to the funds acquired from illegal sources. It is necessary to take into account the crimes of ML by focusing on other sectors that are often used for ML, such as restaurants, hotels and law offices by matching several indicators such as: size of the establishment, the number of workers, the amounts of exports and imports and money transfers as well.

In addition, from region perspective, Saudi Arabia is a member of the FATF group and it was the first Arab country to obtain membership in this group [[Saudi Arabia Full member of the Financial Action Task Force \(FATF\), 2020](#)]. It was observed that Saudi Arabia has not been considered and described by the researchers in this field. Recently, [Naheem \(2019\)](#) published a qualitative research to study Saudi Arabia's practices against ML and terrorist financing; the main goal was to provide an actual profile about AML in Saudi Arabia. For that reason, there is ample scope for future research by using machine learning techniques to identify effective AML tools that would help Saudi Arabia to combat this crime as the European Commission proposed adding Saudi Arabia and another six countries to an ML list because they do too little to thwart the financing of terrorism and organized crime ([Saudi Arabia regrets EU money laundering list proposal - Saudi Gazette, 2020](#)).

The main contribution of this SLR is that it offers a comprehensive analysis of the extant AML literature to avail a high-level view of the state of the art in this domain of research.

References

- About - Financial Action Task Force (FATF) (2020) "[WWW document]", available at: www.fatf-gafi.org/about/ (accessed 28 November 2018).
- Alexandre, C. and Balsa, J. (2015), *A Multiagent Based Approach to Money Laundering Detection and Prevention*, in: ICAART (1). pp. 230-235.
- Alexandre, C. and Balsa, J. (2016), *Integrating Client Profiling in an anti-Money Laundering Multi-Agent Based System*, in: *New Advances in Information Systems and Technologies*, Springer, pp. 931-941.
- Badal-Valero, E., Alvarez-Jareño, J.A. and Pavia, J.M. (2018), "Combining Benford's law and machine learning to detect money laundering. An actual Spanish court case", *Forensic Science International*, Vol. 282, pp. 24-34.
- Colladon, A.F. and Remondi, E. (2017), "Using social network analysis to prevent money laundering", *Expert Systems with Applications*, Vol. 67, pp. 49-58.
- Demetis, D.S. (2018), "Fighting money laundering with technology: a case study of Bank X in the UK", *Decision Support Systems*, Vol. 105, pp. 96-107.
- Dreżewski, R., Dziuban, G., Hernik, Ł. and Pączek, M. (2015b), "Comparison of data mining techniques for money laundering detection system", *2015 International Conference on Science in Information Technology (ICSITech), IEEE*, pp. 5-10.
- Dreżewski, R., Sepielak, J. and Filipkowski, W. (2015a), "The application of social network analysis algorithms in a system supporting money laundering detection", *Information Sciences*, Vol. 295, pp. 18-32.

-
- Friedman, J., Hastie, T. and Tibshirani, R. (2001), *The Elements of Statistical Learning*, Springer series in statistics New York, NY.
- Jayasree, V. and Siva Balan, R. (2017), "Money laundering regulatory risk evaluation using bitmap index-based decision tree", *Journal of the Association of Arab Universities for Basic and Applied Sciences*, Vol. 23 No. 1, pp. 96-102.
- Jullum, M., Løland, A., Huseby, R.B., AAnonsen, G. and Lorentzen, J. (2020), "Detecting money laundering transactions with machine learning", *J. Money Laund. Control*.
- Kannan, S. and Somasundaram, K. (2017), "Autoregressive-based outlier algorithm to detect money laundering activities", *Journal of Money Laundering Control*, Vol. 20 No. 2, pp. 190-202.
- Lawrencica, C. and Ce, W. (2019), "Fraud detection decision support system for Indonesian financial institution", in: *2019 International Conference on Information Management and Technology (ICIMTech)*. Presented at the 2019 International Conference on Information Management and Technology (ICIMTech), IEEE, Jakarta/Bali, Indonesia, pp. 389-394.
- Le Khac, N.A. and Kechadi, M.-T. (2010), "Application of data mining for anti-money laundering detection: a case study", in *2010 IEEE International Conference on Data Mining Workshops*. Presented at the 2010 IEEE International Conference on Data Mining Workshops (ICDMW), IEEE, Sydney, TBD, Australia, pp. 577-584.
- Le-Khac, N.-A. Markos, S. O'Neill, M. Brabazon, A. and Kechadi, T. (2016), "An efficient search tool for an anti-money laundering application of a multi-national bank's dataset", ArXiv Prepr. ArXiv160902031.
- Li, X., Cao, X., Qiu, X., Zhao, J. and Zheng, J. (2017), "Intelligent anti-money laundering solution based upon novel community detection in massive transaction networks on spark", in *2017 Fifth International Conference on Advanced Cloud and Big Data (CBD)*. Presented at the 2017 Fifth International Conference on Advanced Cloud and Big Data (CBD), IEEE, Shanghai, China, pp. 176-181.
- Liu, R., Qian, X., Mao, S. and Zhu, S. (2011), "Research on anti-money laundering based on core decision tree algorithm", in: *2011 Chinese Control and Decision Conference (CCDC)*. Presented at the 2011 23rd Chinese Control and Decision Conference (CCDC), IEEE, Mianyang, China, pp. 4322-4325.
- Moustafa, T.H., Abd El-Megied, M.Z., Sobh, T.S. and Shafea, K.M. (2015), "Anti money laundering using a two-phase system", *Journal of Money Laundering Control*, Vol. 18 No. 3, pp. 304-329.
- Mubalalike, A.M. and Adali, E. (2018), "Deep learning approach for intelligent financial fraud detection system", in: *2018 3rd International Conference on Computer Science and Engineering (UBMK)*, IEEE, pp. 598-603.
- Naheem, M.A. (2019), "Saudi Arabia's efforts on combating money laundering and terrorist financing", *J. Money Laund. Control*.
- Paula, E.L., Ladeira, M., Carvalho, R.N. and Marzagao, T. (2016), "Deep learning anomaly detection as support fraud investigation in Brazilian exports and anti-money laundering", in: *2016 15th IEEE International Conference on Machine Learning and Applications (ICMLA)*, IEEE, pp. 954-960.
- Salehi, A., Ghazanfari, M. and Fathian, M. (2017), "Data mining techniques for anti money laundering", *Int. J. Appl. Eng. Res*, Vol. 12, pp. 10084-10094.
- Sapozhnikova, M., Nikonov, A., Vulfin, A., Gayanova, M., Mironov, K. and Kurennov, D. (2017), "Anti-fraud system on the basis of data mining technologies", in *2017 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT)*, IEEE, pp. 243-248.
- Saudi Arabia Full member of the Financial Action Task Force (FATF) (2020) "[WWW document]", available at: www.sama.gov.sa/en-US/News/Pages/news-355.aspx (accessed 28 November 2019).

-
- Saudi Arabia regrets EU money laundering list proposal - Saudi Gazette (2020) "[WWW document]", available at: www.saudigazette.com.sa/article/559170/SAUDI-ARABIA/Saudi-Arabia-regrets-EU-money-laundering-blacklist-proposal (accessed 30 November 2019).
- Savage, D., Wang, Q., Zhang, X., Chou, P. and Yu, X. (2017), "Detection of money laundering groups: Supervised learning on small networks", in: *Workshops at the Thirty-First AAAI Conference on Artificial Intelligence*.
- Singh, K. and Best, P. (2019), "Anti-money laundering: using data visualization to identify suspicious activity", *International Journal of Accounting Information Systems*, Vol. 34, p. 100418.
- Soltani, R., Nguyen, U.T., Yang, Y., Faghani, M., Yagoub, A. and An, A. (2016), "A new algorithm for money laundering detection based on structural similarity", in: *2016 IEEE 7th Annual Ubiquitous Computing, Electronics and Mobile Communication Conference (UEMCON)*, IEEE, pp. 1-7.
- Suresh, C., Reddy, K.T. and Sweta, N. (2016), "A hybrid approach for detecting suspicious accounts in money laundering using data mining techniques", *International Journal of Information Technology and Computer Science*, Vol. 8 No. 5, p. 37.
- Syed Mustapha Nazri, S.N.F., Zolkafli, S. and Omar, N. (2019), "Mitigating financial leakages through effective money laundering investigation", *Managerial Auditing Journal*, Vol. 34 No. 2, pp. 189-207.
- Villalobos, M.A. and Silva, E. (2017), "A statistical and machine learning model to detect money laundering: an application".
- Yang, S. and Wei, L. (2010), "Detecting money laundering using filtering techniques: a multiple-criteria index", *Journal of Economic Policy Reform*, Vol. 13 No. 2, pp. 159-178.
- Zhang, Y. and Trubey, P. (2018), "Machine learning and sampling scheme: an empirical study of money laundering detection", *Comput. Econ*, pp. 1-21.
- Zhou, Y., Wang, X., Zhang, J., Zhang, P., Liu, L., Jin, H. and Jin, H. (2018), "Analyzing and detecting money-laundering accounts in online social networks", *IEEE Netw*, Vol. 32 No. 3, pp. 115-121.

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